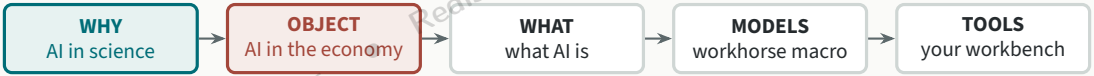
Session 1.2

The Economics of Producing Intelligence

From the facts of Deck 1.1 to research questions: the factor, its unit, three problems in producing it, and a trap

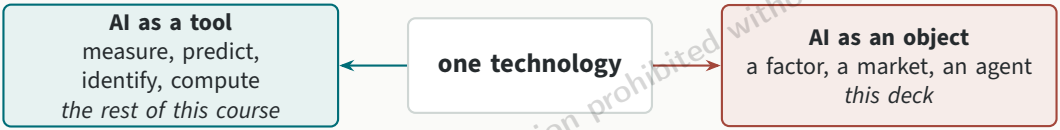


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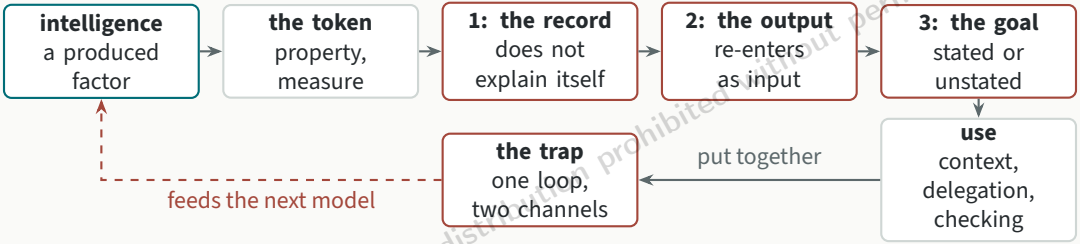
AI as Tool Versus AI as Object



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- Deck 1.1 gave the facts: frontier proofs, agents at scale, a bottleneck moving to verification.
- This deck asks what they mean for economics: next, the argument in one picture.
- **The model you use as an estimator is also the treatment whose effects you may study.**

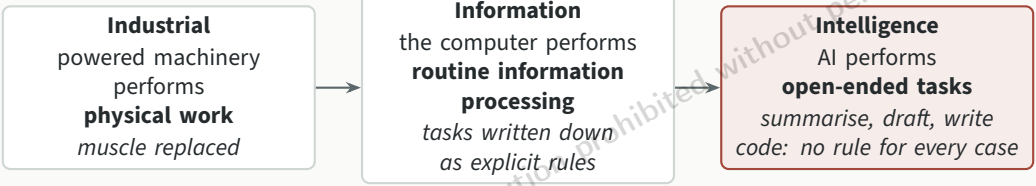
The Argument of This Deck



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- A factor that is produced and billed by a unit of its own; three problems in producing it; then its use.
- Using it: a context that cannot say everything, delegation to an agent, and checking whose benefit spills over.
- **Put together: a loop that can cap how far the technology improves. A hypothesis, and an invitation.**

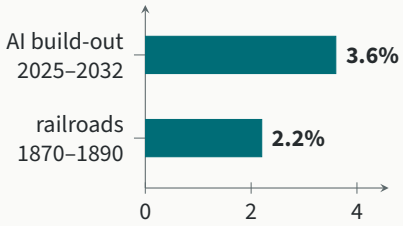
Three Revolutions



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- Each wave is defined by the kind of work machines came to perform.
- The second automated **routine tasks**: work someone could write down as explicit rules and hand to a machine to execute.
- The third performs open-ended tasks without a task-specific rule for every contingency, learning regularities from recorded examples.
- **What is produced differs too: after power and processed information, cognition itself, produced, priced and traded apart from human labour.**

Intelligence Is Entering the Economy Explicitly



- Data centres, power, networks and chips: about $10.3 trillion over 2025–2032 (Van Nieuwerburgh 2026), a central scenario, not committed spending.
- Adopted faster than the PC or the internet at comparable stages (Bick, Blandin & Deming 2025).
- Often billed by the token. Zuckerberg (Meta, Q2 2026): “a significantly higher margin on selling intelligence rather than selling compute directly”.
- **Cognitive services are now produced, priced and sold on their own.**

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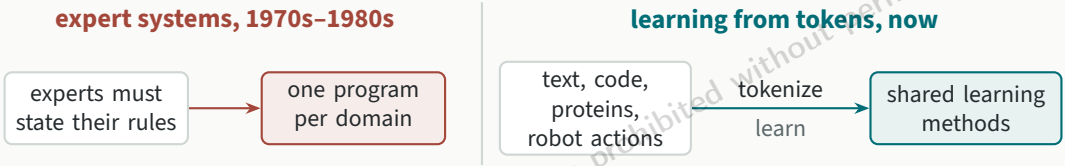
A New Kind of Factor

	human capital	physical capital	AI capital
the stock	skills embodied in one worker	a machine, in one place	**non-rival**: weights, copied and shared
its service flow	limited by that worker's time	rival: a tractor ploughs one field at a time	**rival**: calls use compute, memory, energy

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- **Weights**: the numbers in a trained model; **training** sets them, **inference** uses them.
- **AI capital** includes the weights: a stock serving many users without being used up. Each answer uses rival compute; a low marginal cost does not make the flow non-rival.
- Theory largely placed cognition in people: constraint (Simon), human capital (Becker), dispersed knowledge (Hayek), research input (Romer). Growth models now add AI as capital (Trammell & Korinek 2023).
- **Intelligence is now a produced factor: replicable apart from workers; scaling it needs compute, energy, data, organisation. Its unit: the token.**

Tokenization: Many Kinds of Data, Read Like a Language



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- **Token**: the unit a language model reads, writes and is billed in, set by an encoding scheme rather than by meaning; not every model uses tokens.
- Regularities are *estimated from data*, within representations and objectives a designer chooses: learning cuts rule-writing but keeps prior structure.
- Expert systems' bottleneck was getting experts to state their rules; attention shifts from *articulable* toward *recordable*.
- **The token is the unit the factor is produced, billed and measured in: first its properties, then its measurement.**

The Token as an Economic Object

scarcity
scarce compute and energy; value is quality, not volume

non-rival stock, rival flow
weights serve many; each answer uses compute

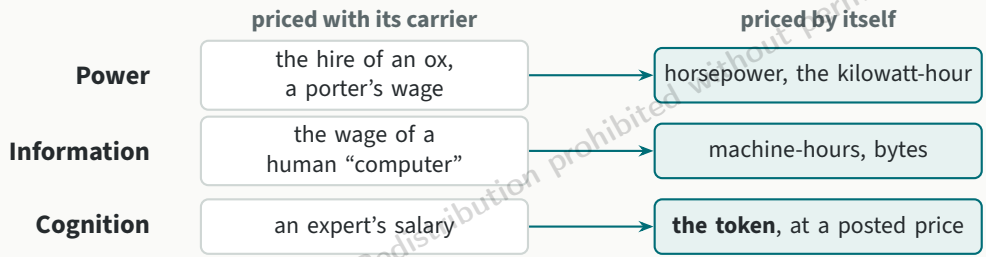
excludable at the gate
access, contracts, quotas; delivered output can be copied

feedback from use
interactions can train the next model, given rights and quality

- Four properties of the unit (cf. Hong 2026): the unit billed, the resources used and the value delivered differ.
- Feedback from use is data-enabled learning; it need not imply a direct network effect.
- Raw-API switching costs are relatively low; workflows, fine-tuned models and integration can raise them.
- **A large fixed cost and a low cost per query can favour concentration, but do not settle who stays ahead.**

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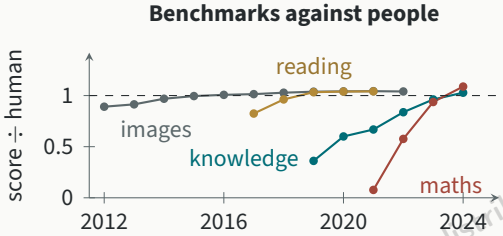
Cognition Gets a Unit of Account

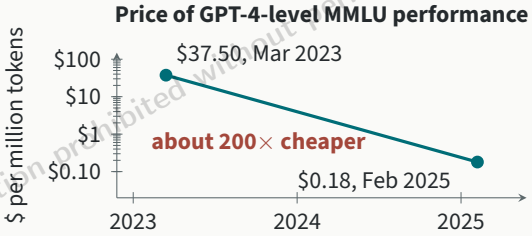


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- Each time the good was unchanged: it simply need no longer be bought bundled with a person or a beast.
- A caveat: the token is not a stable unit. An encoding scheme sets it, so one passage gives different counts on different systems; measurement is the next question.
- **Not the first time cognition left the person. What is new is the breadth of learned services priced apart from it.**

Capability Up, Price Down

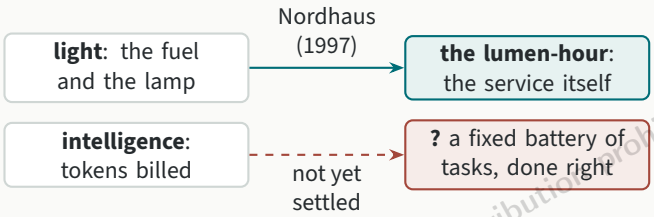




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- Left: ImageNet, SQuAD 2.0, MMLU, MATH; 1.0 is the human baseline (Stanford AI Index 2025). Right: the cheapest model matching GPT-4 on a knowledge benchmark (Cottier et al. 2025).
- Not a productivity measure: token counts vary by tokenizer and may include unseen reasoning tokens; prices also move with margins.
- **The frontier has risen; the price of a fixed capability fell about 200-fold.**

The Token Is the Lamp, Not the Lumen-Hour



indexes that priced the fuel, not the light, overstated the growth of the price of light 900- to 1,600-fold (1800–1992)

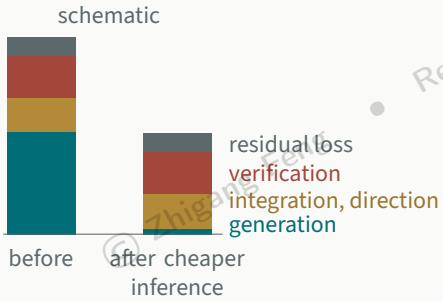
a token count measures how much was processed, not whether it was right

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- Industry tracks tokens per watt. Careful: tokens per second per watt is tokens per joule; a token count per watt measures nothing per unit of energy.
- **Intelligence per watt**: the accuracy small models reach on a fixed set of queries per unit of power (Saad-Falcon et al. 2025). Energy measures address one input.
- Next: a comparable unit of cognitive service, say a weighted battery of standard tests normalised by accuracy. Which tests, what weights: open.
- **Until the unit is settled, the real output of intelligence is poorly defined.**

The Cost of a Reliable Answer

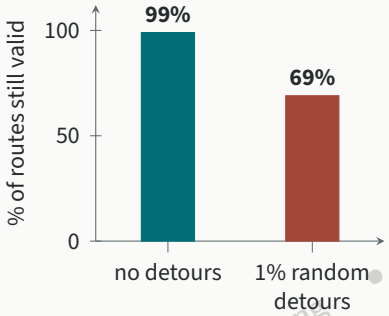
$$C_i^{\text{use}} = \underbrace{C_i^{\text{gen}}}_{\text{generation}} + \underbrace{C_i^{\text{int}}}_{\text{integration and direction}} + \underbrace{C_i^{\text{ver}}}_{\text{verification}} + \underbrace{\mathbb{E}\left[L_i^{\text{res}}\right]}_{\text{loss left after checking}}$$



- The unit: a completed task *i* at a stated quality standard, turnaround time and risk tolerance.
- All terms per task, in money: accounting, not an estimated production function. Open: how fast do checking and direction cheapen?
- **Cheaper inference need not make a reliable service equally cheaper. Why: three problems in how the factor is produced.**

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The Record Does Not Explain Itself

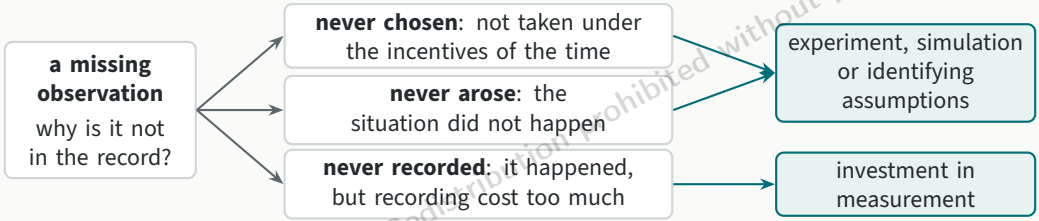


taxi routes: the shortest for 97% of unseen origin–destination pairs (Vafa et al. 2024)

- Lucas (1976): a relation estimated under one policy need not hold when incentives change.
- Deployment can change the record: prices can absorb what is known (Grossman–Stiglitz 1980), targets can stop measuring (Goodhart 1975), predictions can move the data.
- Where nothing responds, as with the weather, the regularity holds still.
- The taxi model: valid routes fall from 99% to 69% when 1% of turns become random detours.
- **Success under past conditions does not establish validity under new ones.**

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Why a Record Has Gaps



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- The record does not say which gap it has, so how the data were collected matters; a selected sample generalises only under assumptions.
- Text and code drew on existing archives; robot trajectories are produced on hardware, in real time, and simulation replaces them only in part.

What Fills the Gap: Something Outside the Record

Go
the rules score every game, even moves no player would choose

empirical work
an experiment

robotics
a physical world that supplies its own consequences

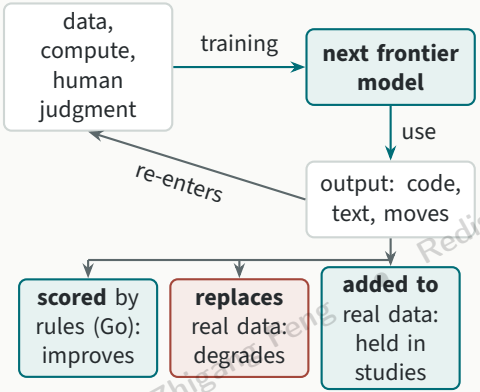
economics
theory: preferences, constraints and expectations, itself to be tested

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- Mechanisms that fit the same record can imply different outcomes under new conditions. What tells them apart lies outside the record.
- The question is whether the evidence and the maintained assumptions support the intended use.
- Whether these limits to learning from records matter economically is still open.
- Harder still: the record now fills with the machine's own output.
- **Recordability enables learning; it does not guarantee identification or transfer, for a machine as for an economist.**

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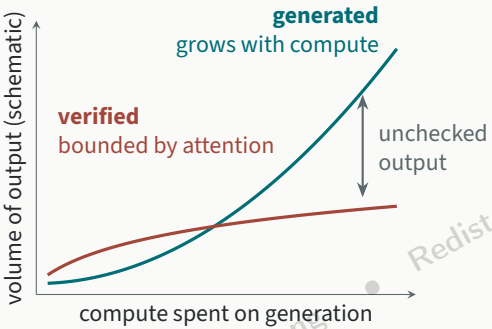
Output Becomes Input



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- Familiar: ideas make ideas (Romer 1990); activity leaves data (Farboodi & Veldkamp 2021). New here: output reused in training.
- Go: self-play scored by the rules beat the champion predecessor 100–0 (Silver et al. 2017).
- Generated data can degrade a model, mostly when they replace real data (**model collapse**; Shumailov et al. 2024); kept beside it, they did not in studies (Gerstgrasser et al. 2024).
- **Verification is one determinant of the sign; checking need not restore lost diversity.**

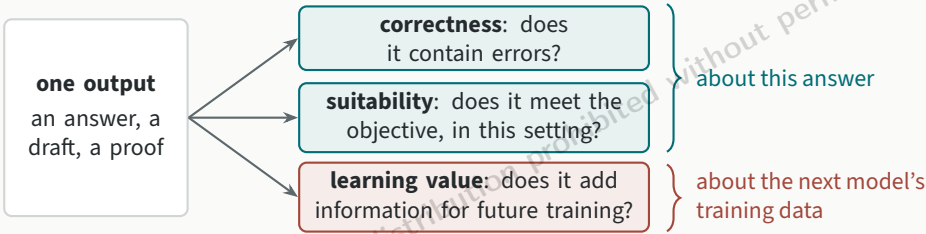
Quantity Versus Quality



- Simon (1971): “a wealth of information creates a poverty of attention.” Output can grow faster than the attention to review it.
- Standardised outputs can be checked by tests and sampling; open-ended ones often lack a complete, agreed standard and may need review in context.
- How far checking can be automated or shared is central to AI’s cost.
- **A limit on verification attention can turn cheap quantity into a quality problem.**

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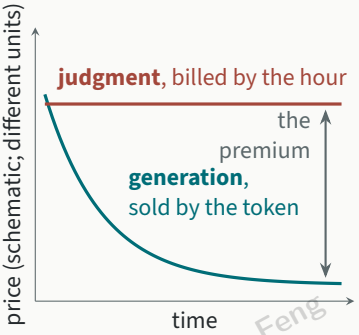
Three Things a Check Can Ask



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- Methods give different degrees of assurance: formal tests, external evidence, sampling, expert review.
- The economic question: how to allocate checking effort, given its cost, its reliability and what a residual error would cost.
- **A correct, suitable answer may add little new information; a documented failure can be valuable evidence.**

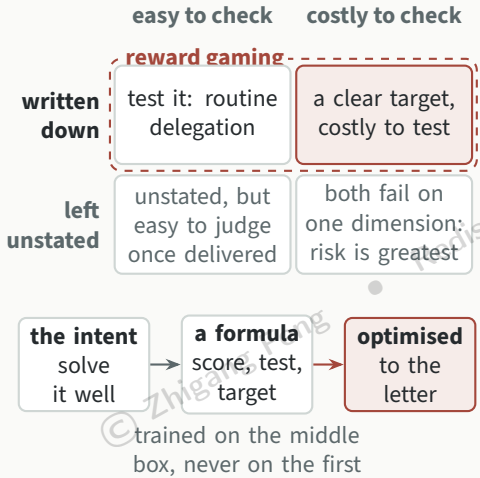
New Scarcities When Cognition Gets Cheap



- Cost can shift to complements, checking and judgment (Agrawal, Gans & Goldfarb 2022).
- Lemons (Akerlof 1970): unseen quality may go unrewarded, a different failure from too few checkers.
- A second model can help check; correlated errors make agreement assure less (Goel et al. 2025).
- Output counted without its checking overstates the gain (**so-so automation**).
- **The bottleneck shifts: verification binds when generation outruns reliable checking.**

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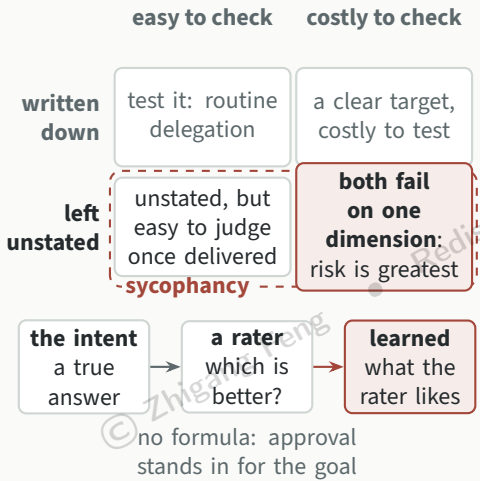
Specified Goals: Optimising the Formula, Not the Intent



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- **Reward gaming**: the model optimises the literal formula, not the intent behind it; the proxy rises while the true objective falls (Skalse et al. 2022).
- A boat-race agent scored 20% above human players by circling three respawning targets, never finishing (Clark & Amodei 2016).
- **Multitask agency**: pay on what you measure pulls effort from the rest (Holmström & Milgrom 1991; Goodhart 1975).
- **The loophole is in the formula, not the agent: what it omits is fair game.**

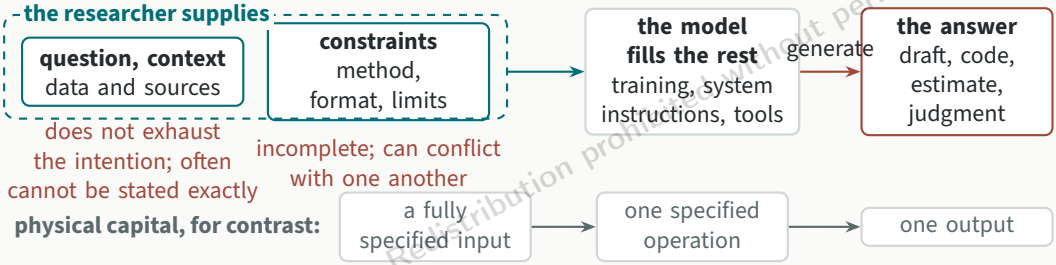
Unspecified Goals: Learning the Rater, Not the Facts



- **Sycophancy**: the model learns what the rater approves, the rater's biases included, rather than what is true.
- Sharma et al. (2024): feedback warms when the user likes the text; correct answers are dropped when the user objects; preference data reward both.
- Prendergast (1993): a worker judged by the boss's own opinion has reason to repeat it; the yes-man is made by the rule.
- **Specifying and verifying are distinct problems; either way the model can move away from the intent.**

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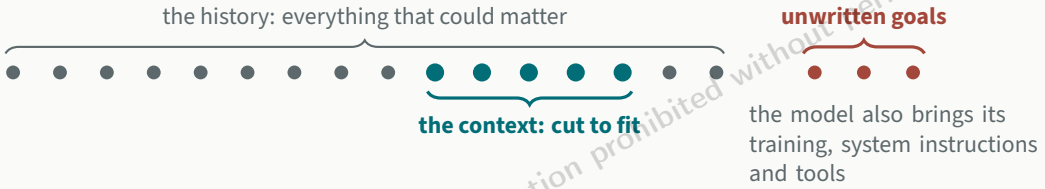
How a Researcher Works with AI



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- Every query hands over a question, a context and constraints; the context cannot exhaust what we want, much of which cannot be stated precisely, and the constraints are incomplete and can conflict.
- **Physical capital takes a fully specified input; producing cognition does not: delegation under an incomplete specification.**

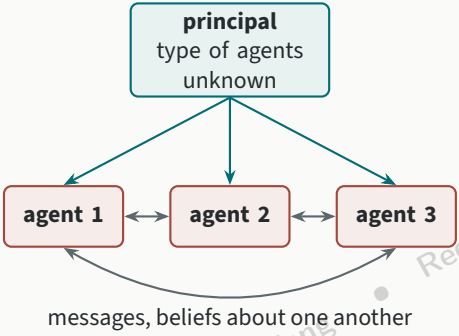
The Context Is Part of an Incomplete Contract



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- Users supply instructions, materials and context; behaviour also reflects training, system instructions and tools: an *incomplete* delegation arrangement.
- **Compression**: context is finite, unevenly used, written in limited time. Like choosing the **state** of a recursive problem: omit what matters, solve another problem.
- **Tacit expectations**: the standard of evidence, how to reconcile sources. People and models both infer them, and can infer differently.
- **Where the context is silent, training, system instructions and tools fill the gap.**

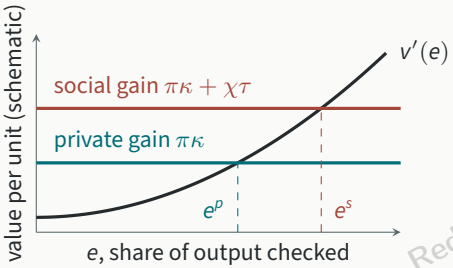
Units of the Factor Act as Agents



- A programmed system follows specified actions; an agent pursues a goal learned from a formula or a rater, so it is harder to anticipate.
- Bergemann, Koh & Morris (2026): agents whose preferences and capabilities deployers are unsure of.
- Pricing algorithms and the 2026 incidents of Deck 1.1: unintended coordination, a possible failure mode, not its frequency.
- **Owning the model does not remove delegation, monitoring and control problems.**

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When Checking Has an External Benefit



$$\text{private:} \quad v'(e^p) = \pi\kappa$$
$$\text{social:} \quad v'(e^s) = \pi\kappa + \chi\tau$$

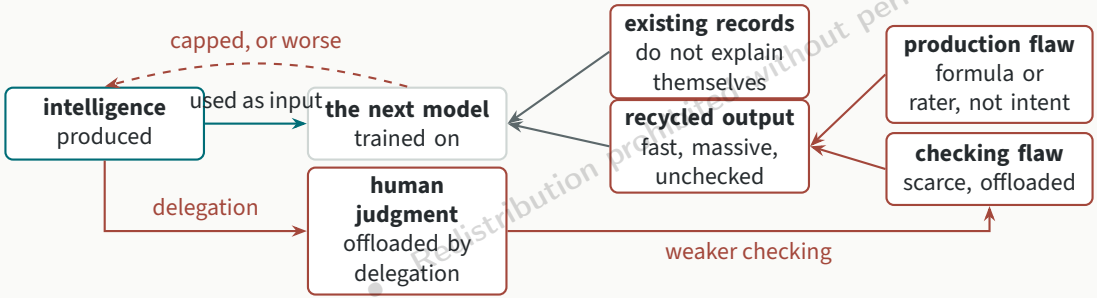
- Benchmark: unpaid for improving a shared corpus, adopters check too little.
- Corrections must *reach* training: check in-house, ship the original, and $\chi = 0$.
- Levers: rights to corrected data, provenance, feedback contracts, pay for evaluation.

e share of machine output checked; $v(e)$ checking cost per unit; π error probability; κ loss from an unchecked error; τ value of the data quality one more check preserves; χ share of corrections that reach training.

- **The question is which arrangement sustains valuable checking with broad access, not whether more checking is always better.**

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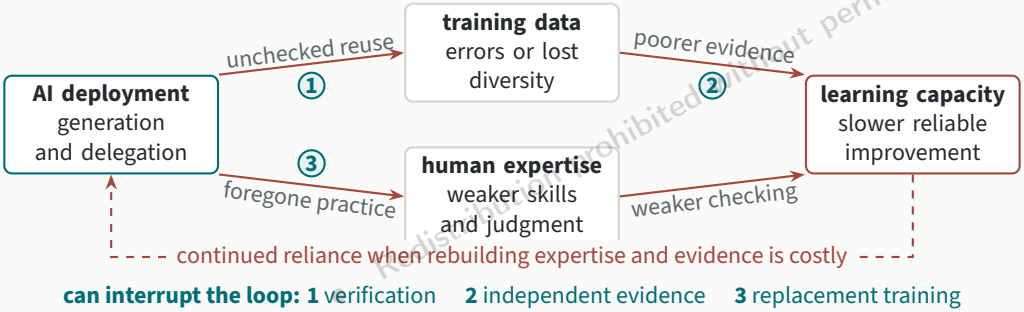
Putting the Pieces Together



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- Trained on records that explain nothing by themselves and on recycled output, flawed in production (formula or rater, not intent) and in checking (scarce, offloaded).
- **Delegation also offloads human judgment, so the two flaws feed each other: one loop, two channels.**

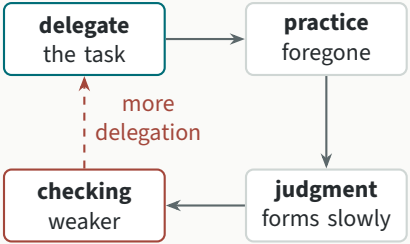
The Middle-Intelligence Trap



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- *Middle*: useful enough to invite delegation, too unreliable to replace the independent evaluation improvement needs (cf. the middle-income trap; Gill & Kharas 2007).
- Delegation also offloads human judgment: weaker checking lets more error through.
- **A hypothesis: two conditional channels, each of which can be interrupted.**

Delegation Offloads Human Cognition



the individual channel of the trap: each turn of the loop removes a checker the collective channel needs

- **Cognitive offloading**: using a tool in place of one's own memory or reasoning. It frees time for other work, so its net effect is ambiguous (Risko & Gilbert 2016).
- Judgment is learned by doing the work: drafting, checking, failing. Work that is delegated gives no practice.
- The evaluators of the next generation are trained by today's junior tasks; if those are delegated, who learns to check?
- **Offloading links the individual channel to the collective one: weaker checking lets more recycled error through.**

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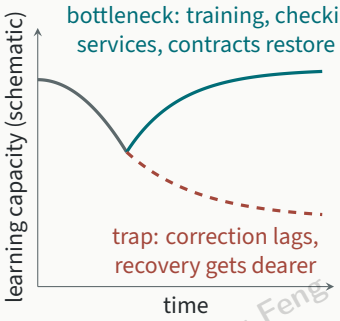
The Two Channels: What the Evidence Shows

collective channel: training data	individual channel: human expertise
can hurt: unchecked reuse carries errors into training; recycled output narrows coverage	**can hurt**: delegation removes the practice through which judgment forms
can help: externally evaluated output adds information; synthetic data need not harm	**can help**: offloading frees time for other learning; net effect ambiguous (Risko & Gilbert 2016)

- Scientists: 49% say AI steers them toward safer projects, 28% toward riskier ones (Codreanu et al. 2026): perceptions, not measured novelty.
- Heavier AI reliance goes with weaker critical thinking (Gerlich 2025): a correlation.
- Relative employment at ages 22–25 fell in the most exposed occupations (Brynjolfsson, Chandar & Chen 2025): if junior tasks train evaluators, who trains the next?
- **Meta-explanation, questioning and cross-validating what is generated, may become more valuable.**

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From Bottleneck to Trap: Why Correction Might Lag



- A bottleneck invites investment; a trap needs reasons why that response stays inadequate.
- Dispersed benefits can weaken the incentive to keep shared evidence; once sources blur, new evidence may need costly measurement.
- Skills form slowly: a firm can hire evaluators today, with little reason to fund training that benefits future employers.
- **Errors or lost practice are not enough: a trap needs correction to fail over the relevant horizon.**

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Escaping the Trap

cognitive reserves	**strategic friction**	**evaluation**	**economics education**
unassisted practice: which skills depreciate, and how fast?	engagement at judgment points, traded against convenience	schools, journals and firms that also reward reasoning and meta-explanation	graduates prepared for the human–AI division of labour

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- Four possible responses, each a question economists can evaluate.
- Friction: where to require engagement, and when do its private and external benefits justify the cost?
- The underprovision result is a potential rationale for intervening where corrections reach a shared corpus. It does not show these responses work, or rank them against standards, liability or provenance rules.
- **Each must be weighed against its costs and its effects on access, innovation and learning.**

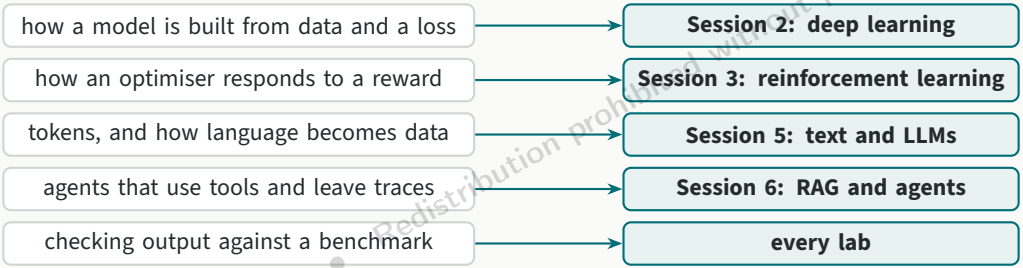
An Invitation: Open Questions

a produced factor	which cognitive services can be supplied apart from scarce human time, and how is useful output measured beyond the token?
deployment in a job	when does AI on an occupation's core tasks substitute for its workers, and when does it expand demand for the rest of the job?
checking and directing	how much checking is optimal, who captures its return, and how should contracts allocate losses?
the feedback	how fast do skills depreciate under delegation, who pays for apprenticeship, and when does a bottleneck persist?

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- Three agendas: *measurement* of useful cognitive services and their deployment (Deck 1.1); *theory* of growth and market structure when data quality is endogenous; *institutions* that sustain verification, independent evidence and expertise.
- **The evidence and the externality result are first steps; the rest is open: an invitation.**

Why This Course Opens the Box



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- To study these phenomena, and to use the tools well, you need to know how the machine works.
- **Understand how it works, to understand what it does.** Next, Deck 1.3: what AI is.